(iii) $\frac{d^2y}{dx^2} = \sqrt{y - \frac{dy}{dx}}$

(iv) $\frac{d^3y}{dx^3} = 0$

(v) $\frac{d^2y}{dx^2} + y + \left(\frac{dy}{dx} - \frac{d^3y}{dx^3}\right)^{3/2} = 0$

(vi) $\left(2 - y''\right)^2 = y''^2 + 2y'$

(vii) $\left(\frac{dy}{dx}\right)^3 + y = x - \frac{dx}{dy}$

2. Find the differential equation of the following

(i) $y = cx + c - c^3$ (ii) $y = c\left(x - c\right)^2$

(iii) $xy = c^2$ (iv) $x^2 + y^2 = a^2$

3. Form the differential equation by eliminating α and β from $\left(x - \alpha\right)^2 + \left(y - \beta\right)^2 = r^2$.

4. Find the differential equation of the family of all straight lines passing through the origin.

5. Form the differential equation that represents all parabolas each of which has a latus rectum $4a$ and whose axes are parallel to the x axis.

6. Find the differential equation of all circles passing through the origin and having their centers on the y axis.

7. Find the differential equation of the family of parabola with foci at the origin and axis along the x-axis.

Solution of a Differential Equation:

The relation between the dependent and independent variables not involving derivatives is called the solution of the differential equation.

Solution of the differential equation must contain the same number of arbitrary constants as the order of the equation. Such a solution is called General (complete) solution of the differential equation.

4.2 First order and first degree differential equations

A differential equation of first order and first degree can be written as $f\left(x, y, \frac{dy}{dx}\right) = 0$. Here we will discuss the solution of few types of equations.

4.2.1 General solution and particular solution

For any differential equations it is possible to find the general solution and particular solution.

4.2.2 Differential Equation in which variables are separable

If in an equation it is possible to collect all the terms of x and dx on one side and all the terms of y and dy on the other side, then the variables are said to be separable. Thus the general form of such an equation is

$f(x)dx = g(y)dy$ (or) $f(x)dx + g(y)dy = 0$

By direct integration we get the solution.

Example 4.6

Solve: $(x^2 + x + 1)dx + (y^2 - y + 3)dy = 0$

Solution:

Given $(x^2 + x + 1)dx + (y^2 - y + 3)dy = 0$

It is of the form $f(x)dx + g(y)dy = 0$

Integrating , we get

$$\int (x^2 + x + 1)dx + \int (y^2 - y + 3)\, dy = c$$

$$\left(\frac{x^3}{3} + \frac{x^2}{2} + x\right) + \left(\frac{y^3}{3} - \frac{y^2}{2} + 3y\right) = c$$

Example 4.7

Solve $\frac{dy}{dx} = e^{x-y} + x^2e^{-y}$

Solution :

Given $\frac{dy}{dx} = e^{x-y} + x^2e^{-y} = e^{-y}e^{x} + e^{-y}x^2$

$= e^{-y}\left(e^{x} + x^2\right)$

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